As shown in Fig.10.1, substances present in X will tend to cross the boundary between the two liquid layers, to form a dynamic equilibrium.

10.3 Types of chromatography

There are several types of chromatography which are classified based on the interaction of the sample with the mobile and stationary phase. In general, the different chromatographic techniques fall under two major categories: Partition chromatography and adsorption chromatography. Partition chromatography involves partition between two liquids. The stationary phase is a liquid and is held in an inert supporting liquid. In adsorption chromatography, the stationary phase is a finely divided adsorbent such as silica gel and the mobile phase can be a gas or more commonly a liquid. The different forms of chromatography are represented in Fig 10.2

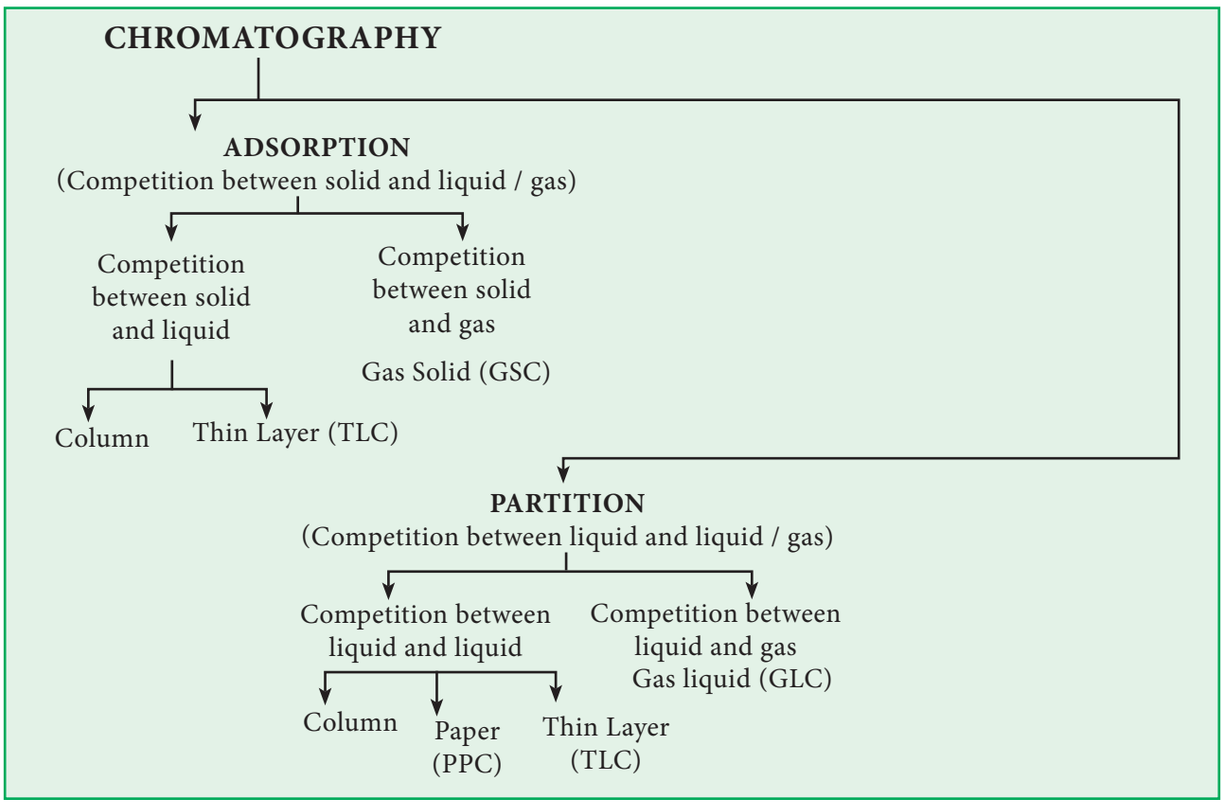

Figure.10.2. Types of Chromatography

10.3.1 Column chromatography

In column chromatography, a mixture of substances, in solution is applied to a column filled with a permeable solid matrix immersed in a solvent. As shown in Fig.10.3, sample, for example, a mixture of protein is applied to the column matrix (packed into a glass or metal surface). After equilibration, the proteins are eluted out using a specific

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